F. Yet Another Problem About Pairs Satisfying an Inequality

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: binary search, data structures, dp, greedy, sortings | https://codeforces.com/problemset/problem/1703/F

You are given an array $a_1, a_2, \ldots a_n$. Count the number of pairs of indices $1 \le i, j \le n$ such that a_i lt i lt a_j lt j.

Input

The first line contains an integer t ($1 \le t \le 1000$) — the number of test cases.

The first line of each test case contains an integer n ($2 \le n \le 2 \cdot 10^5$) — the length of the array.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($0 \le a_i \le 10^9$) — the elements of the array.

It is guaranteed that the sum of n across all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the number of pairs of indices satisfying the condition in the statement.

Please note, that the answer for some test cases won't fit into 32-bit integer type, so you should use at least 64-bit integer type in your programming language (like `long long` for C++).

Example

Input

```
5
8
1 1 2 3 8 2 1 4
2
1 2
10
0 2 1 6 3 4 1 2 8 3
2
1 1000000000
3
0 1000000000 2
```

Output

```
3
0
10
0
1
```

Note

For the first test cases the pairs are $(i, j) = \{(2, 4), (2, 8), (3, 8)\}$.

- The pair (2, 4) is true because $a_2 = 1$, $a_4 = 3$ and 1 lt 2 lt 3 lt 4.
- The pair (2, 8) is true because $a_2 = 1$, $a_8 = 4$ and 1 lt 2 lt 4 lt 8.
- The pair (3, 8) is true because $a_3 = 2$, $a_8 = 4$ and 2 lt 3 lt 4 lt 8.